AMCS506:Energy Equations

Lecture 2

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Chapter 2: 1st and 2nd Laws of thermpdynamics

For the CFD, we need to know mainly four laws as follows:

(iii) The first law of thermodynamics

(iv)The second aw of thermodynamics

The first law of thermodynamics :Energy equation

The energy equation is derived from the ***first law of thermodynamics***, which states that the rate of change of energy of a fluid particle is equal to the rate of heat addition to the fluid particle plus the rate of work done on the particle:

Energy equation:

$$\rho C_p \frac{DT}{\partial t} = div(kgard\ T) = \frac{\partial}{\partial x}(k\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k\frac{\partial T}{\partial y}) + \frac{\partial}{\partial z}(k\frac{\partial T}{\partial z}) + S_T$$

The second law of thermodynamics : The second law leads to a thermodynamic property—*entropy*. For any *reversible* process that a system undergoes during a time interval *dt*, the change in the entropy *S* of the system. In CFD, the entropy is generated by the fluid friction, temperature gradient as follows:

$$S_F = \mu\{2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2\}$$

$$S_T = k\{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2\}$$

Additional laws:

In addition to the general laws, it is usually necessary to bring certain particular laws into analysis. Examples are (a) Fourier's law of heat conduction, (b) Newton's law of cooling, (b) the Stefan–Boltzmann law of radiation, (c) Newton's law of viscosity, and (d) the ideal gas law.

(a)Fourier's law of conduction: $q'' = -k\dfrac{\partial T}{\partial n}$

$q''$is the magnitude of the heat flux in n –direction and $\frac{\partial T}{\partial n}$ is the temperature gradient in the same direction

(b) Newton's law of cooling:$q'' = h(T_w - T_\infty) = -\frac{k\left(\frac{\partial T}{\partial n}\right)_w}{T_w - T_\infty}(T_w - T_\infty)$

(c) Stefan–Boltzmann law of radiation :$q_r'' = \sigma T^4$

Where σ is the Stefan–Boltzmann constant

σ = 5.6697 × 10−8 W/(m2 K4)

(d) Newton's law of viscosity: $\tau = \mu\frac{\partial u}{\partial n}$

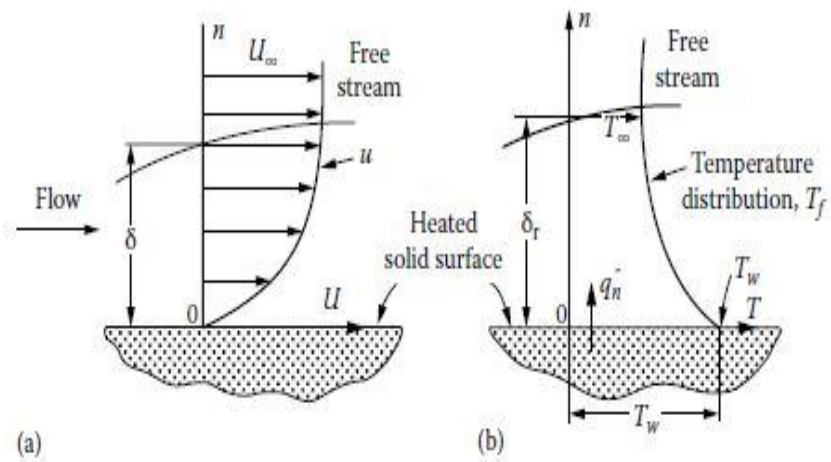


Fig:(a) velocity boundary layer (b) thermal boundary layer

(e)Equation of the state: We can describe the state of a substance in thermodynamic equilibrium by means of just two state variables. ***Equations of state*** relate the other variables to the two state variables. If we use ρ and T as state variables we have state equations for pressure P and specific internal energy i: $P = P(\rho, T)$ and $i = i(\rho, T)$. For a ***perfect gas*** the following, well-known, equations of state are useful:$PV = nRT, P = \rho RT$ and $i = C_v T$. N is the number of moles of a substance, R is the gas constant

Some important non-dimensional numbers in CFD

Reynolds number: Re $= \frac{UL}{\mu/\rho} = \frac{UL}{\nu} = \frac{(inertia\ force)}{(viscosu\ force)}$, Prandtl number: Pr $= \frac{\nu\ (momentum\ diffusivity)}{\alpha\ (thermal\ diffusivity)}$ Where $\alpha = \frac{k}{\rho C_p}$

Grashof number: Gr $= \frac{g\beta\Delta T L^3 (buoyancy\ force)}{\nu^2 (viscosu\ force)}$, Rayleigh number: Ra $= \frac{g\beta\Delta T L^3 (buoyancy\ force)}{\alpha\nu (viscosu\ force)}$,

Lewis number: Le $= \frac{\alpha (thermal\ diffusivity}{D (mass\ diffusion\ rate)}$

Schimdt number: Sc $= \frac{\nu\ (viscosu\ diffusion\ rate)}{D\ (mass\ diffusion\ rate)}$, Ra= PrGr, Richardson number: Ri $= \frac{Gr}{Re^2}$

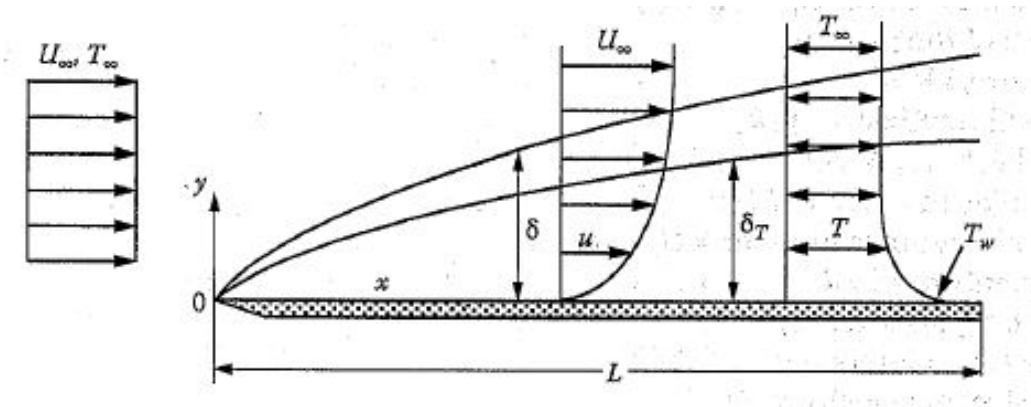
(a) Local Nusselt number : $Nu = -\frac{k\left(\frac{\partial T}{\partial n}\right)_w}{T_w - T_\infty}$

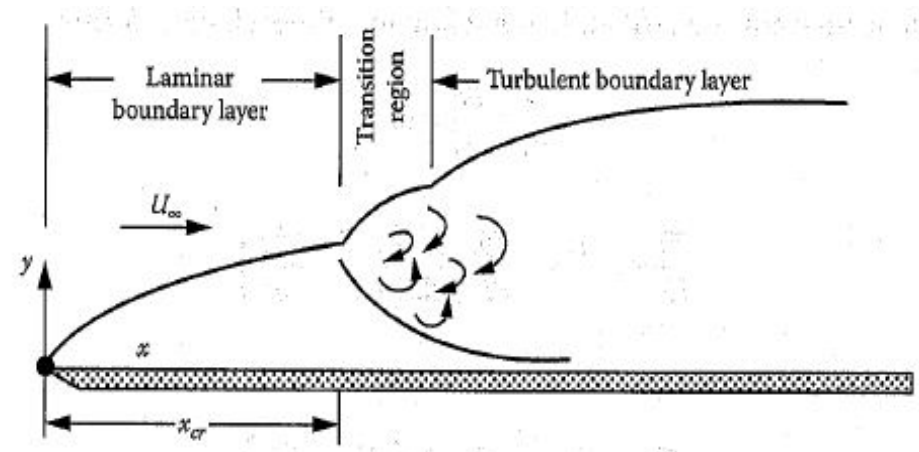
(b) Local Sherwood number :$Sh = -\frac{D\left(\frac{\partial C}{\partial n}\right)_w}{C_w - C_\infty}$

Osborne Reynolds popularised the concept.

State of flows for the internal extremal layer:

(a) Laminar flow (b) transitional flow, and (b) turbulent flow





(a) Laminar boundary layer (b) laminar, transitional and turbulent boundary layer flow

Laminar flow boundary layer equations was discovered by Ludwig Prandtl (1875-4953):

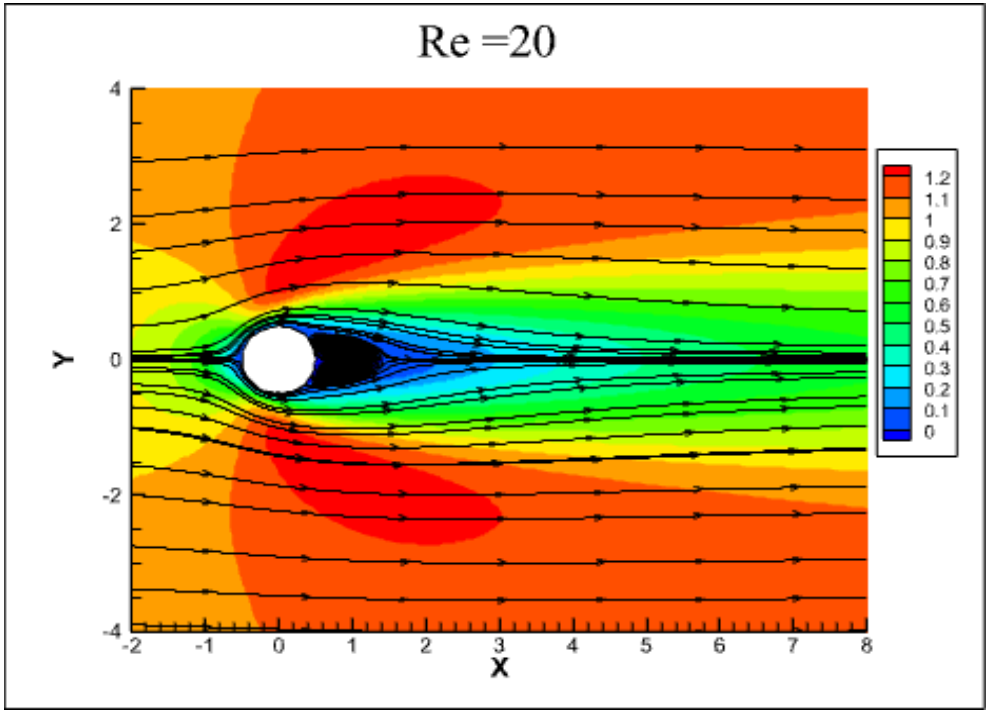
$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0,$$

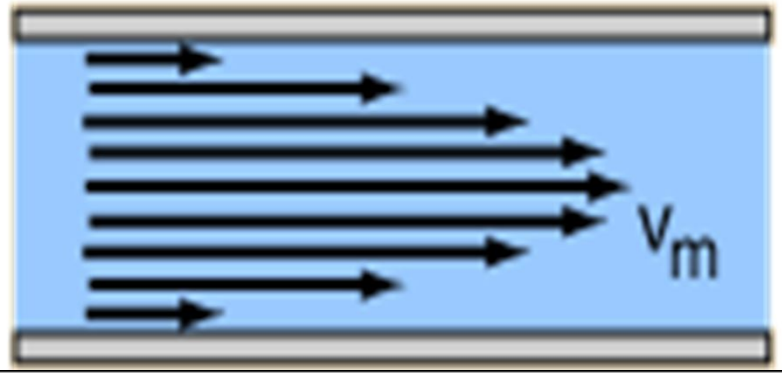
$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{1}{\rho}\frac{dp}{dx}+v\left(\frac{\partial^2 u}{\partial y^2}\right),$$

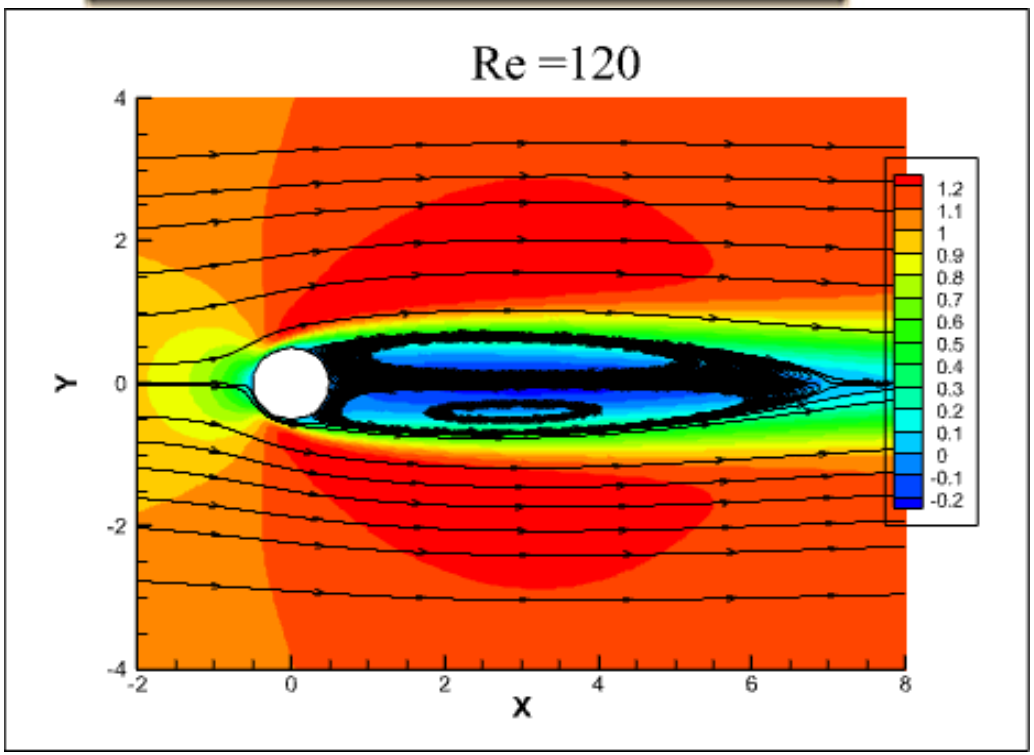
$$u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}=\frac{k}{\rho c_p}\frac{\partial^2 T}{\partial y^2}.$$

Some examples o laminar flows:

- Occurs when a fluid flows in parallel layers, with no disruption between the layers.
- There are no cross-currents perpendicular to the direction of flow, nor eddies or swirls of fluids.







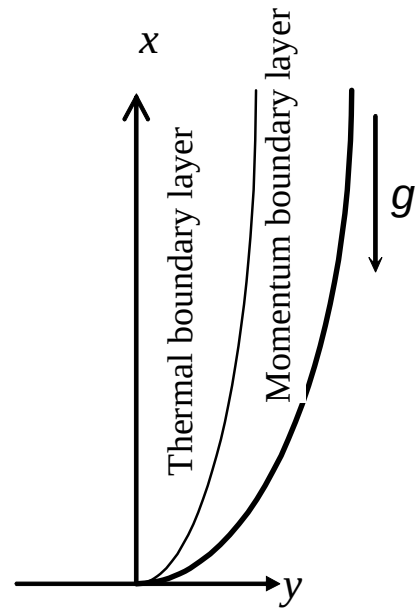
Example: Radiation effects on natural convection with streamwise surface temperature

Boundary

$$\frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} = 0$$

$$\rho\left(\hat{u}\frac{\partial \hat{u}}{\partial \hat{x}} + \hat{v}\frac{\partial \hat{u}}{\partial \hat{y}}\right) = \mu\frac{\partial^2 \hat{u}}{\partial \hat{y}^2} + \rho g \beta\left(T - T_\infty\right)$$

$$\hat{u}\frac{\partial T}{\partial \hat{x}} + \hat{v}\frac{\partial T}{\partial \hat{y}} = \frac{k}{\rho C_p}\frac{\partial^2 T}{\partial \hat{y}^2} - \frac{1}{\rho C_p}\frac{\partial q_r}{\partial \hat{y}}$$



$$q_r = -\frac{4\sigma}{3k\left(\alpha_r + \sigma_s\right)}\frac{\partial T^4}{\partial \hat{y}}$$

Rosseland diffusion approximation (1936) for the radiative heat flux

Boundary conditions

$$\hat{u} = \hat{v} = 0, \quad T = T_\infty + \left(T_w - T_\infty\right)\left[1 + \alpha \sin\left(\pi \hat{x}/l\right)\right] \text{ at } \hat{y} = 0$$

$$\hat{u} \to 0, \quad T \to T_\infty \quad \text{as} \quad \hat{y} \to \infty$$

Results: Radiation effects

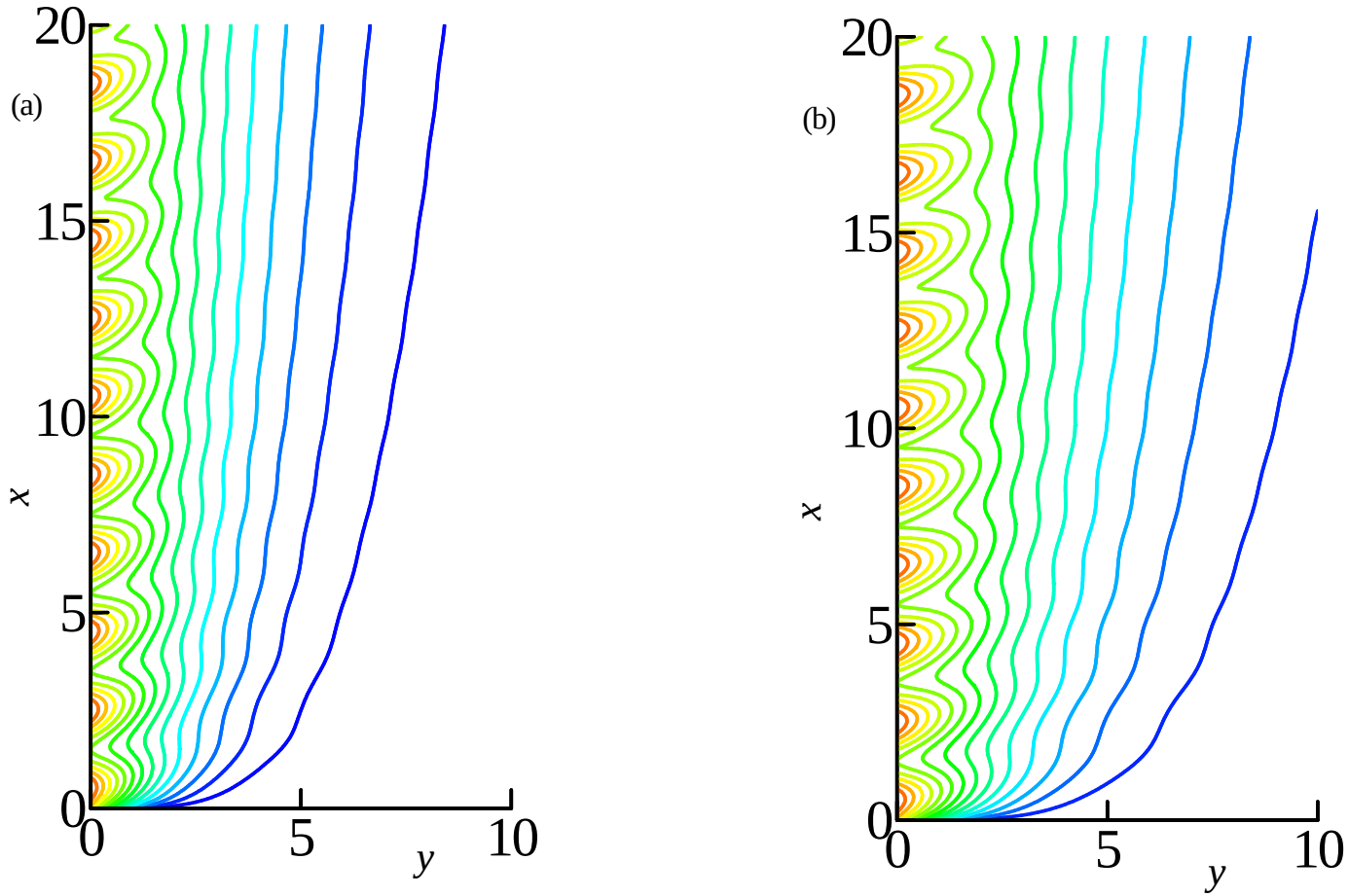
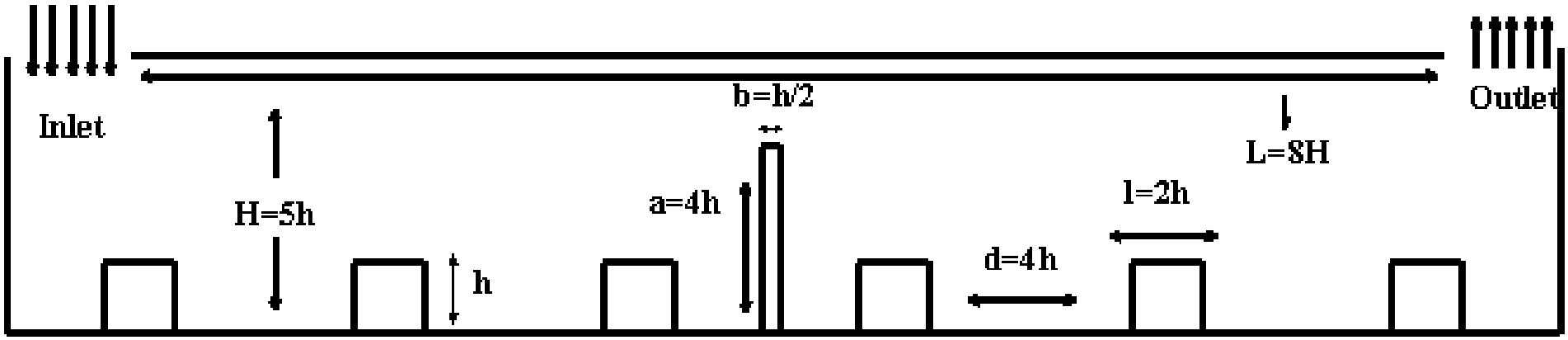


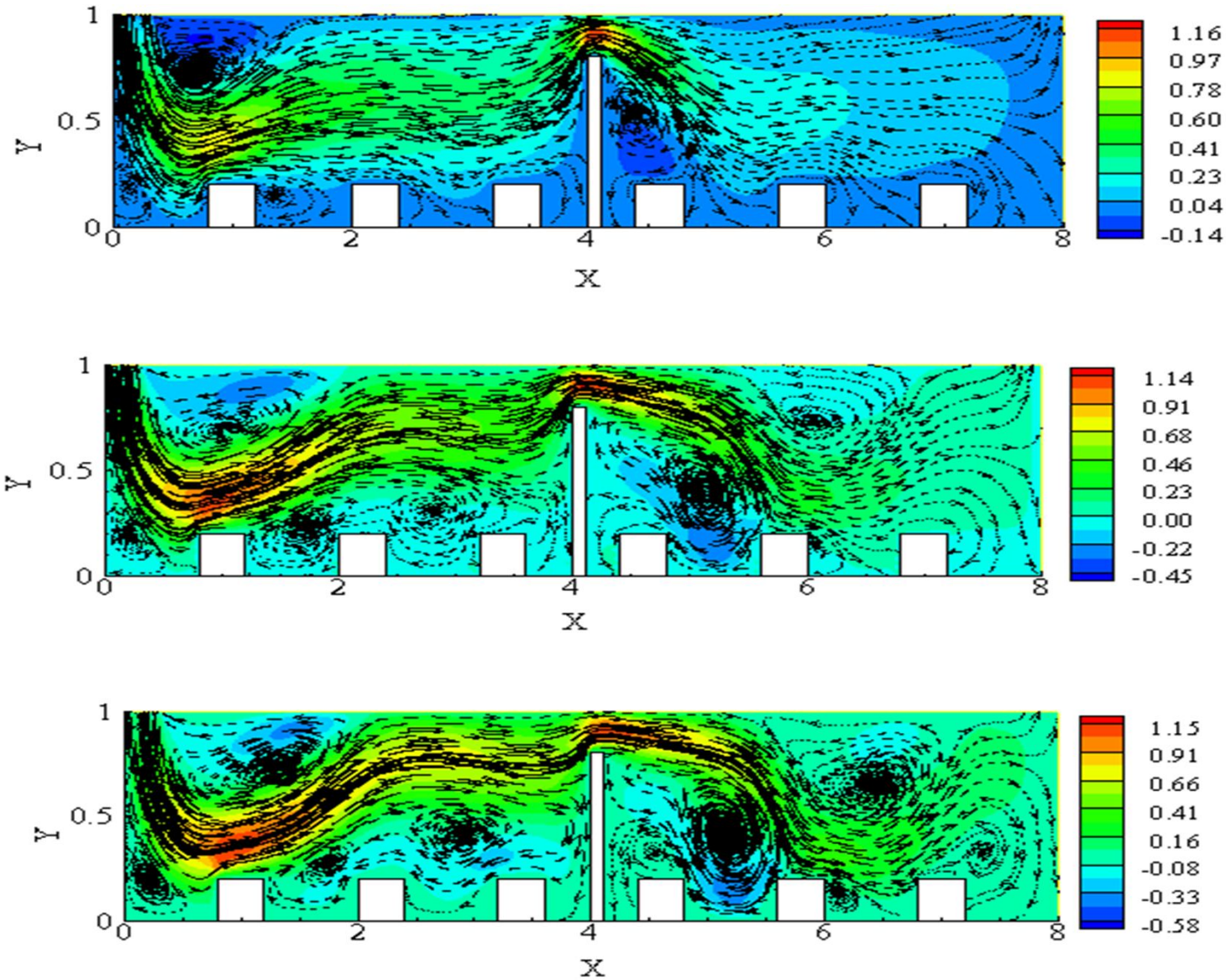
Figure: Isotherms for (a) Rd = 0.0 (b) Rd = 0.5 while Pr = 0.7, α = 0.2

Schematic Diagrams of the hospital



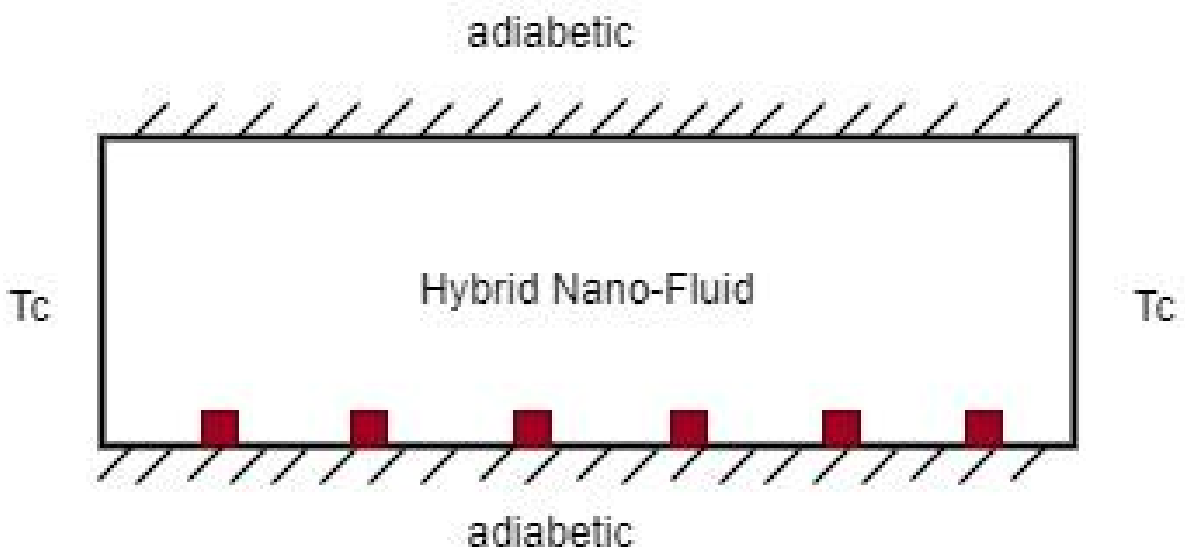
- Room's lattice size: 400 x 50
- Inlet at 0-0.0375 lattice and outlet at 0.9625-1 lattice
- Each block (bed) is of 0.05 x 0.2 lattice.
- Partition is of 0.0125 x 0.8 lattice

Laminar flow



Re =500, 700 and 1000

Convective Heat Transfer Simulation of Nanofolds in a Rectangular Enclosure with Multiple Chips



Analysis for 6 microchips

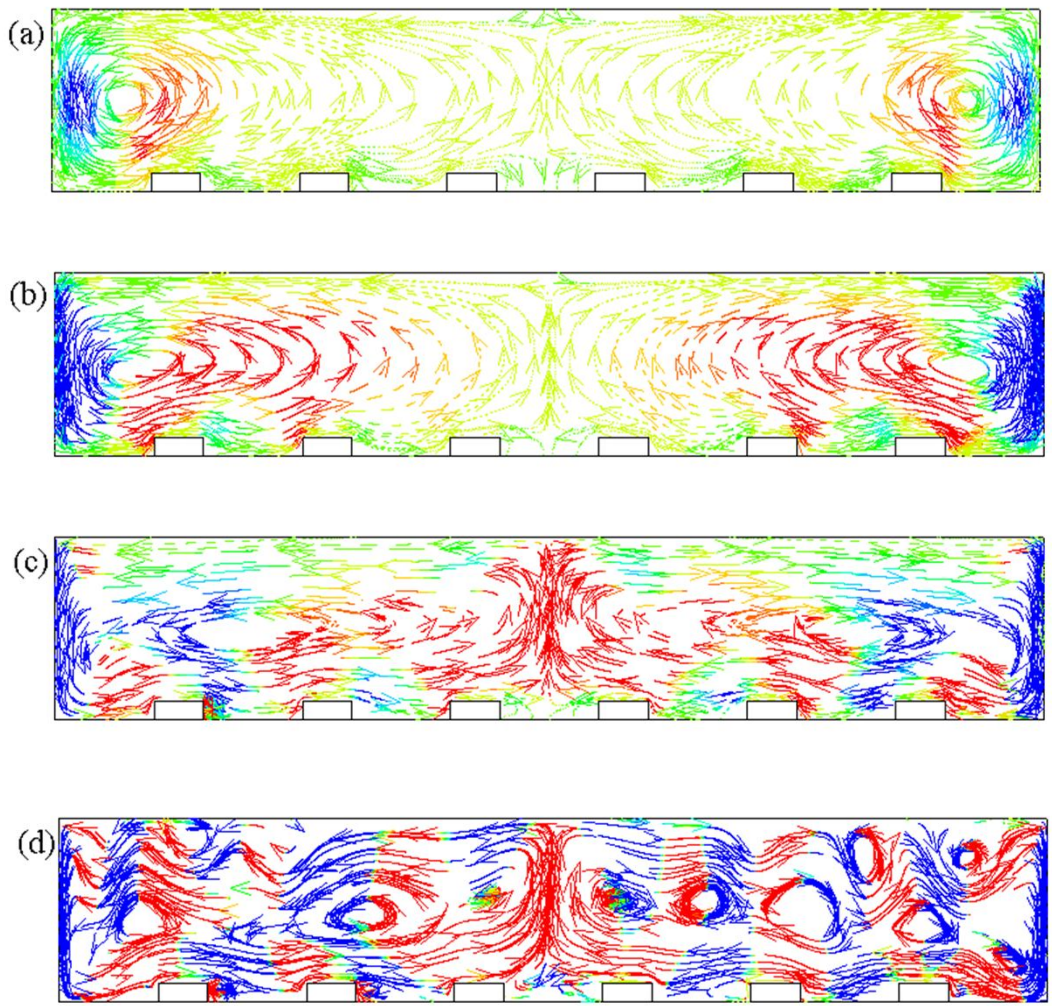


Fig: Streamlines for the four different Rayleigh numbers (a) Ra=10^3 *(b) Ra=*10^4 *(c) Ra=*10^5 *(d) Ra=*10^6

Laminar Simulation of convective heat transfer

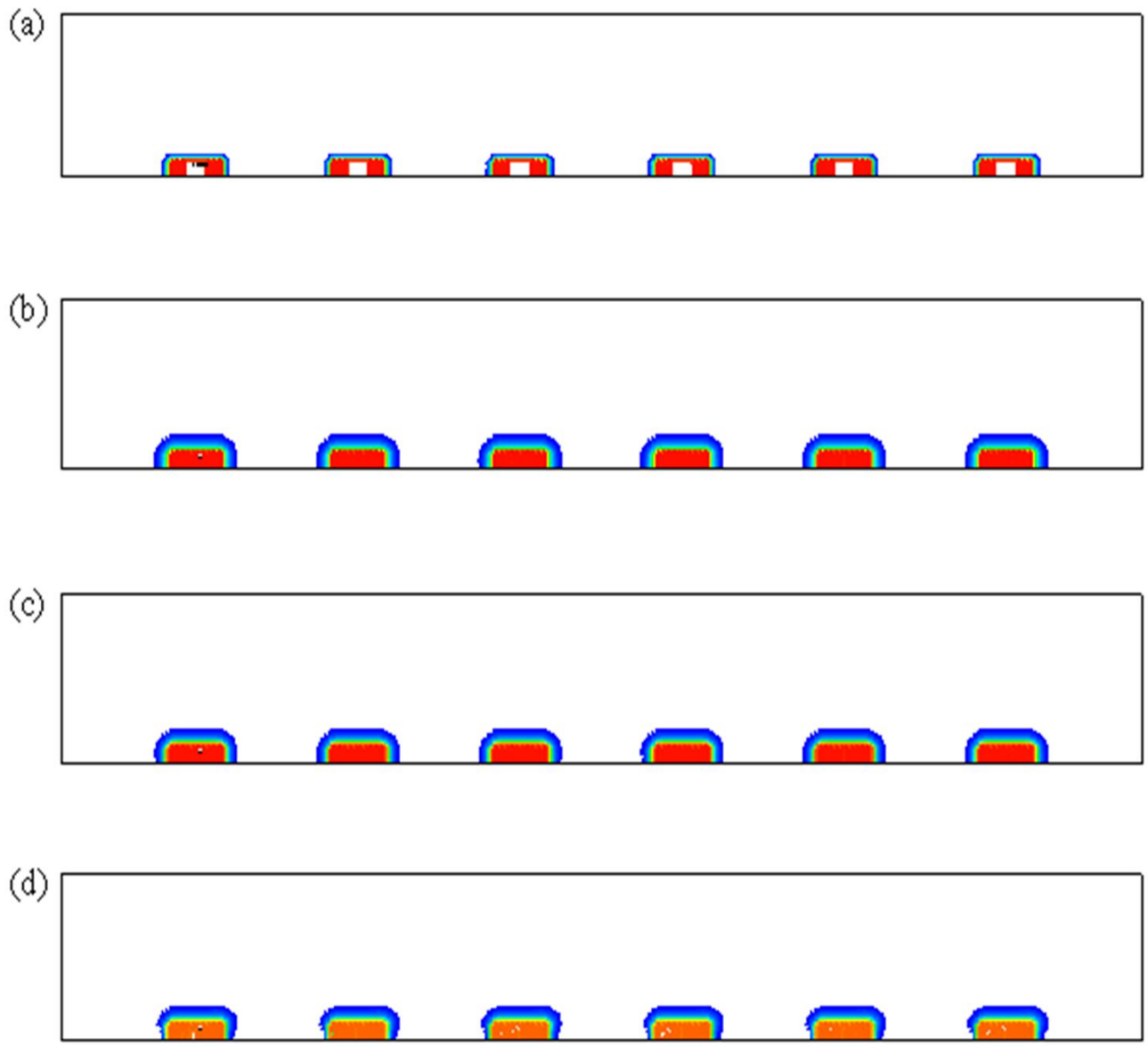


Fig: Isotherms for the four different Rayleigh numbers

(a) $Ra=10^3$

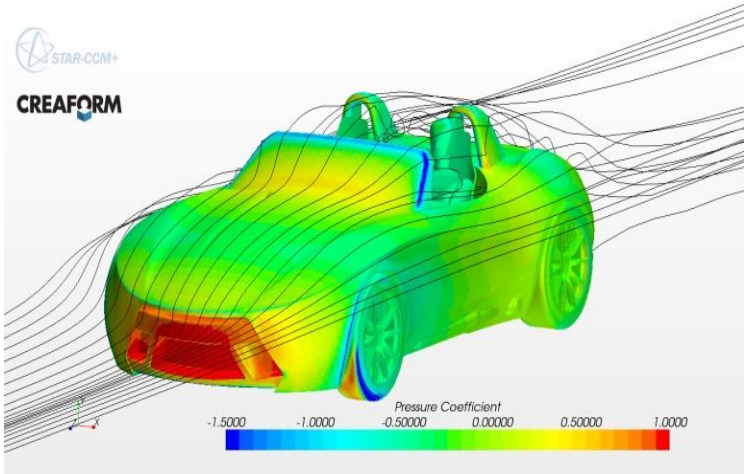
(b) $Ra=10^4$

(c) $Ra=10^5$

(d) $Ra=10^6$

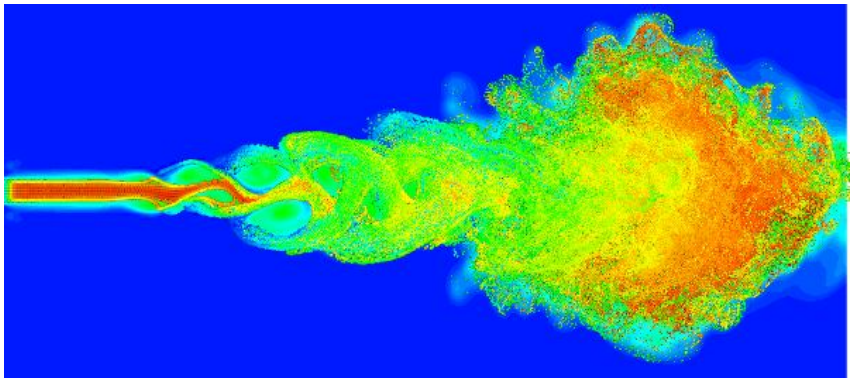
Turbulent flow

- A flow regime characterized by chaotic property changes.
- Unsteady vortices appear on many scales and interact with each other.
- Nobel Laureate Richard Feynman described turbulence as ‘the most important unsolved problem of classical physics’.



An example of CFD

-Google images



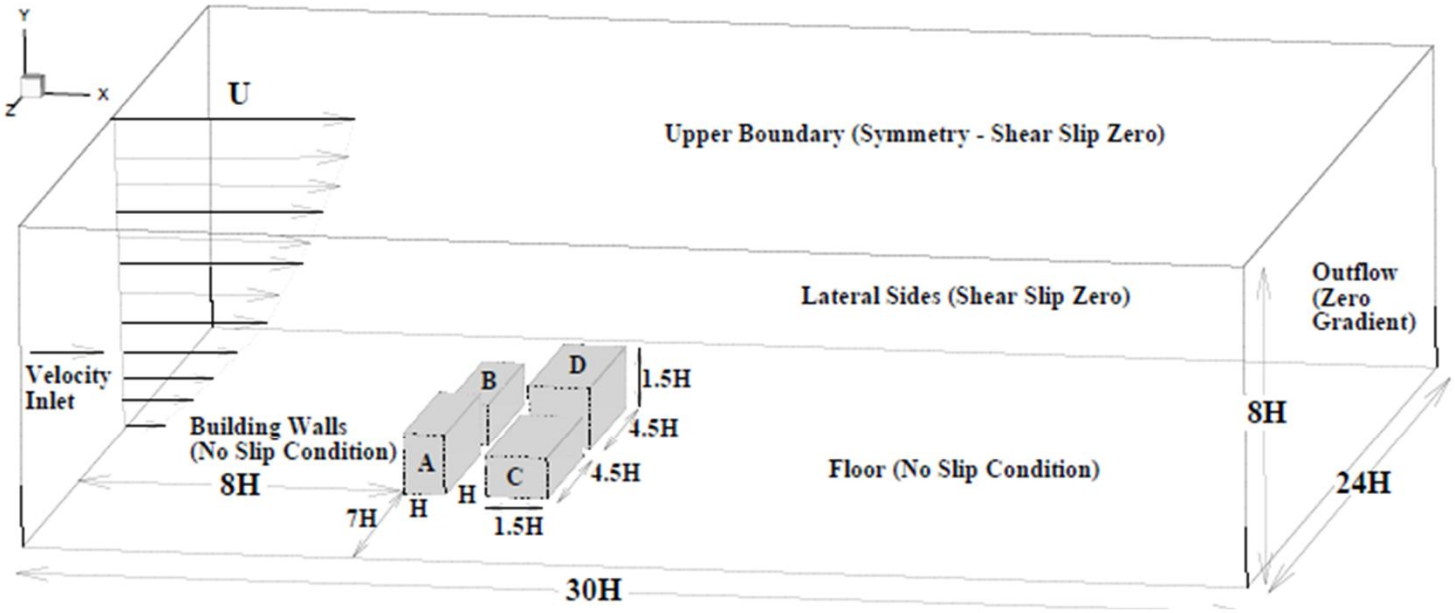
An example of CFD

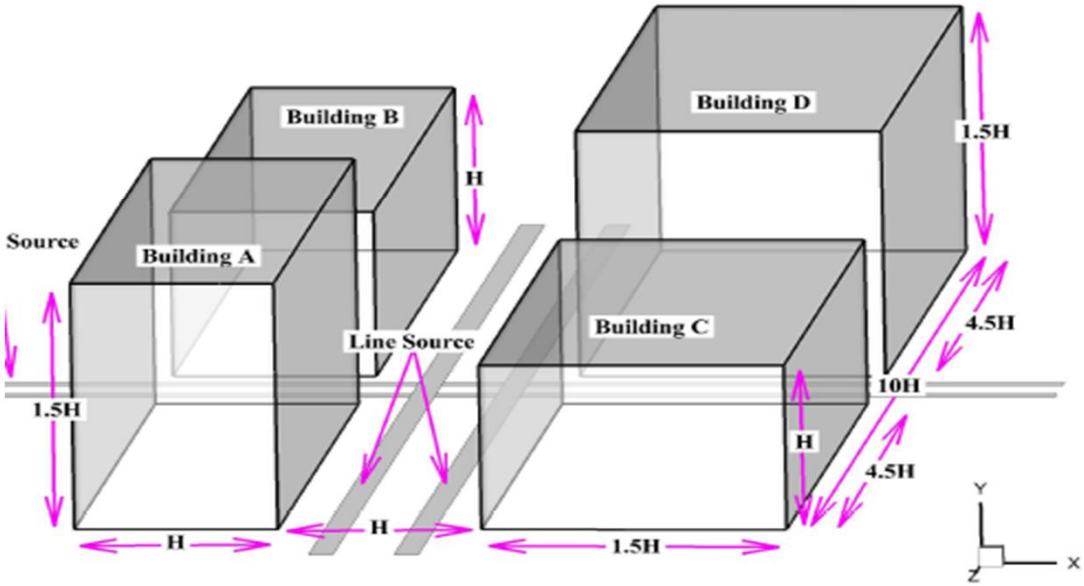
-Google images

Air Pollution and Its Dispersion

- Air pollution is one of the manmade problems in our environment
- It is an atmospheric condition with various substances are present at high level of concentration
- The concentration of substances become higher enough which lead to harmful effects in a certain locality
- The main poisonous substance are: carbon monoxide (CO), Nitrogen oxides (NOx), Sulphur di oxide (SO2), suspended particulate matter (PM-10) and (PM-2.5) etc.
- Pollutant dispersion in different urban areas deals with air turbulence, a complex structure of air flow with different topography (e.g high-rise buildings, street canyons)
- Pollutant dispersion also deals with thermal effect, transport and diffusion of heavy gas or dusty particles in air

Model-2: Wind flow and Pollutant dispersion around a street canyon intersection





(a) Building Distribution.

Model-2: Wind flow and Pollutant dispersion around a street canyon intersection

Numerical (LES) results for 3D streamlines with pollutant concentration:

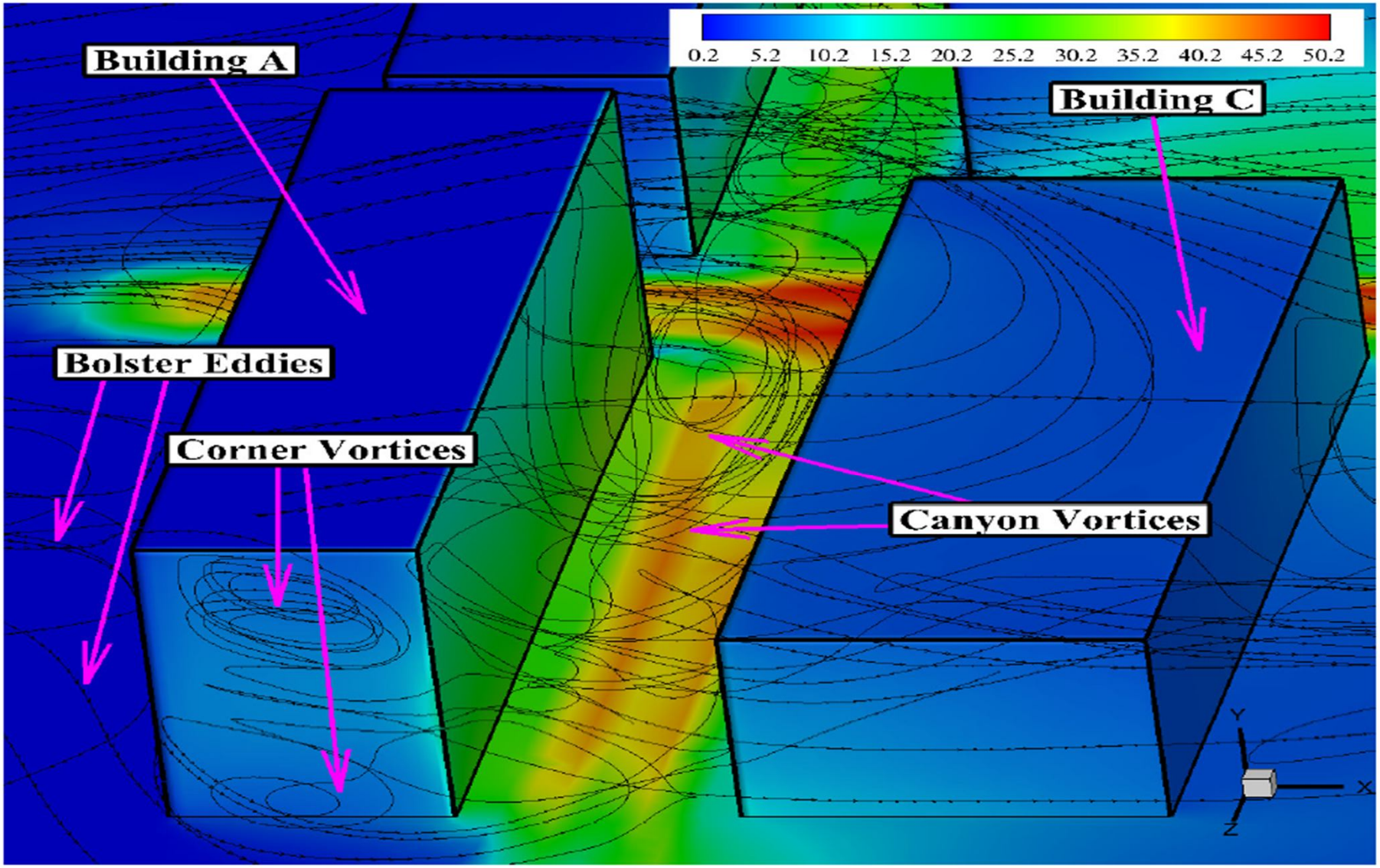


Figure 16. The 3*D* streamlines of time-averaged normalized velocity magnitude $\langle \overline{U} \rangle / u_H$ and contour of time-averaged normalized pollutant concentrations $\langle \overline{C}^+ \rangle$ around Building A and Building C.

Model-2: Wind flow and Pollutant dispersion around a street canyon intersection

Numerical (LES) results for 3D streamlines with pollutant concentration:

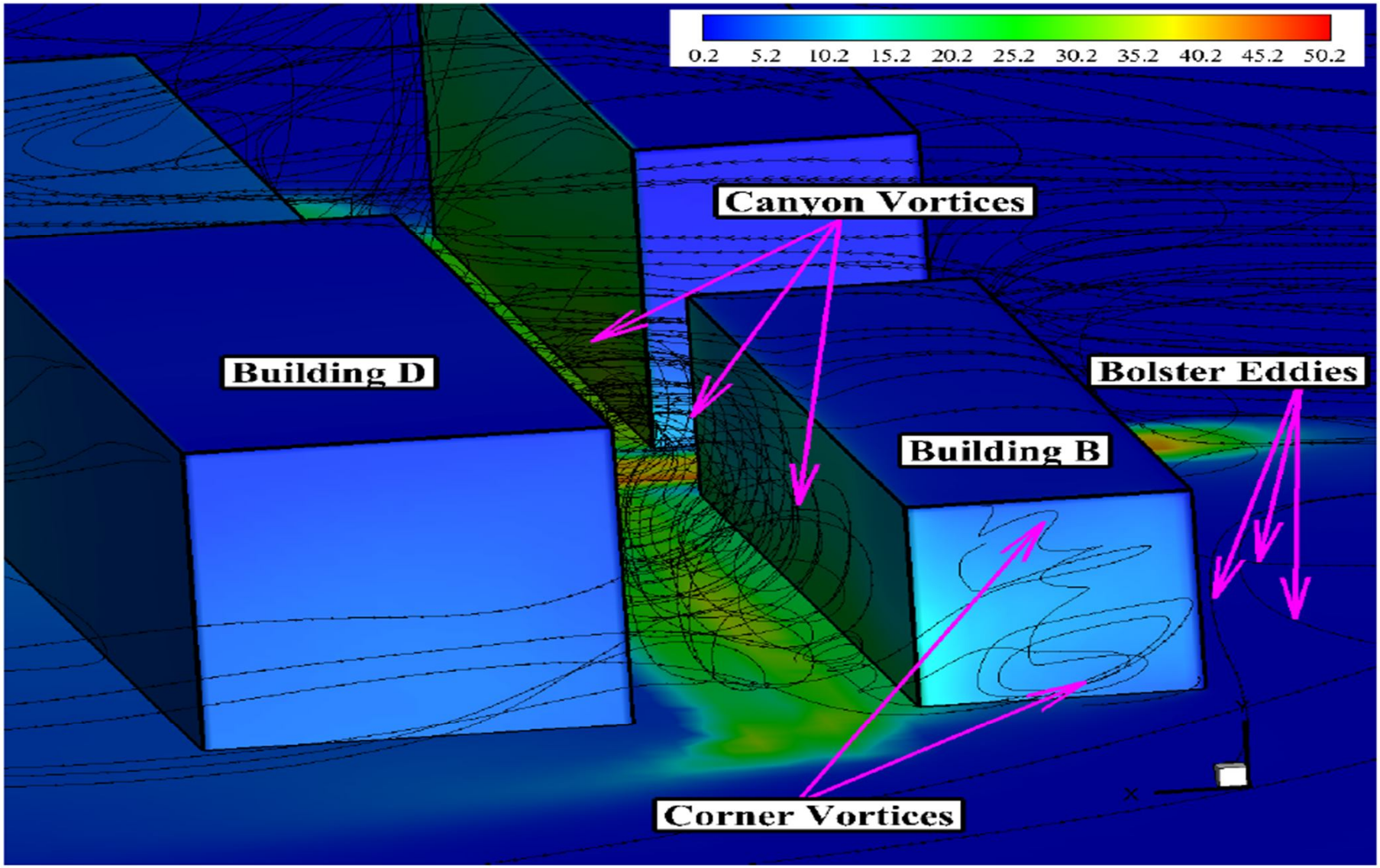


Figure 17. The $3D$ streamlines of time-averaged normalized velocity magnitude $\langle \overline{U} \rangle / u_H$ and contour of time-averaged normalized pollutant concentrations $\langle \overline{C}^+ \rangle$ around Building B and Building D.

Examples of turbulent flow: Blood Flow in a model Arterial Stenosis

- Arterial stenosis is a narrowing of an artery
- High cholesterol deposited inside the blood vessel wall causes plaques
- This plaque retards the blood motion
- As a result blood flow becomes transitional to turbulent

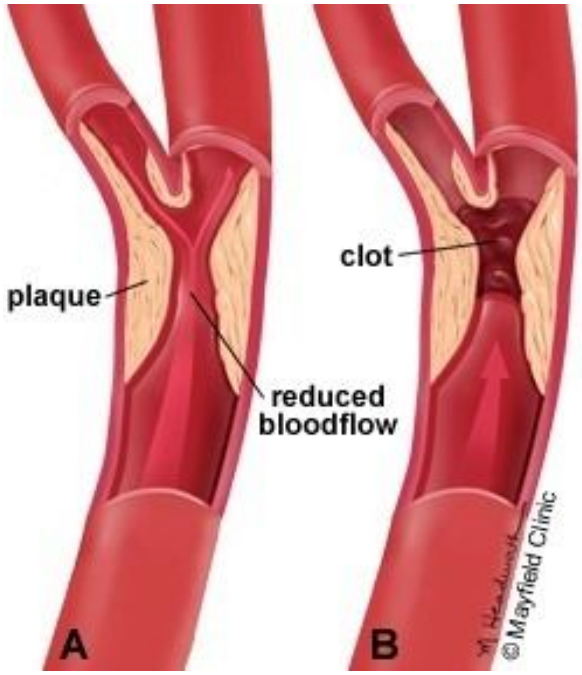


Fig. Arterial stenosis

Courtesy: Google image

Large-eddy Simulation of Transition to Turbulent Pulsatile Flow using a Dynamic Nonlinear Subgrid-scale(SGS) Stress Model

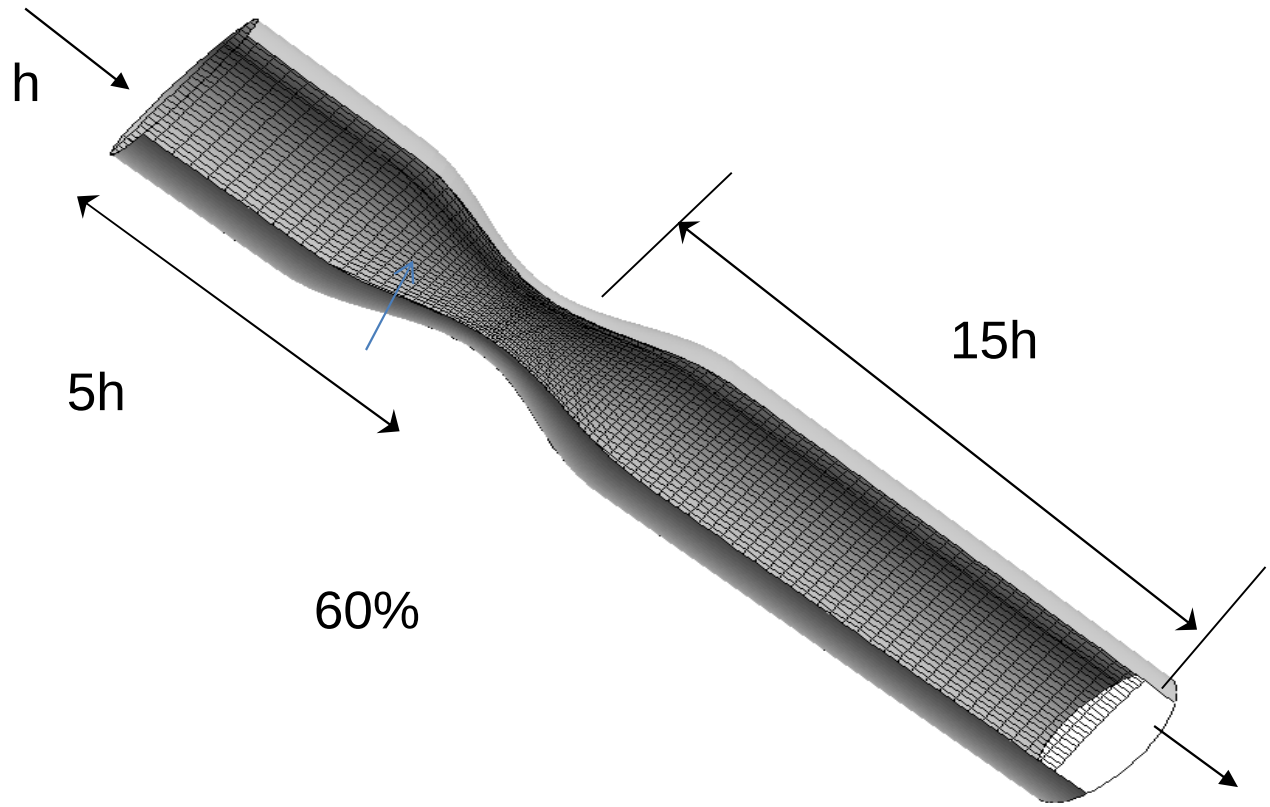
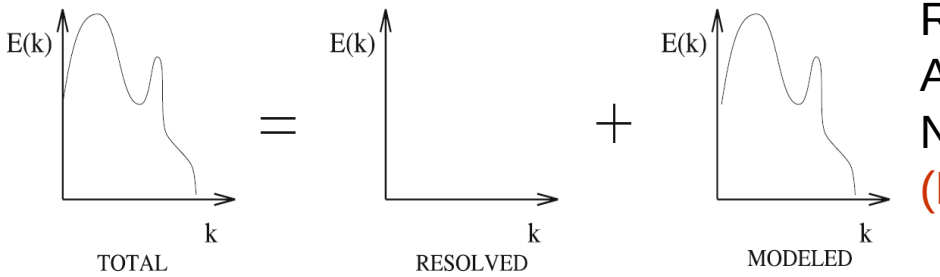
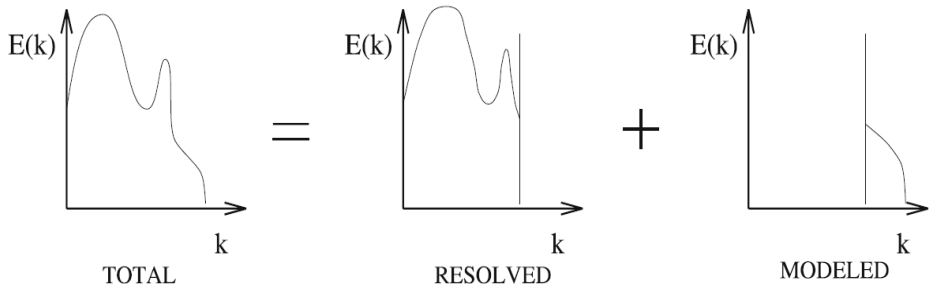


Figure: Geometry of the Physical Model

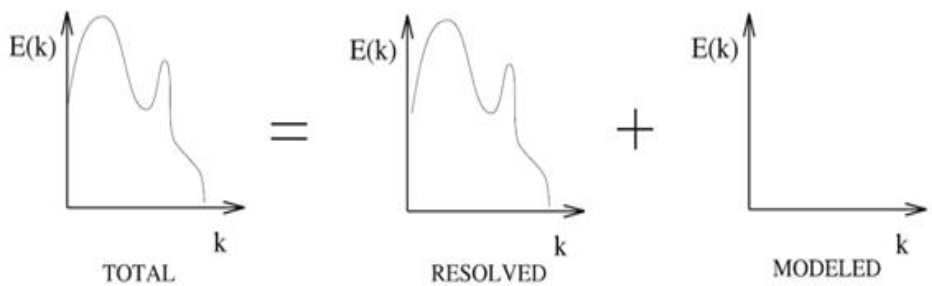
Numerical Simulation Methods



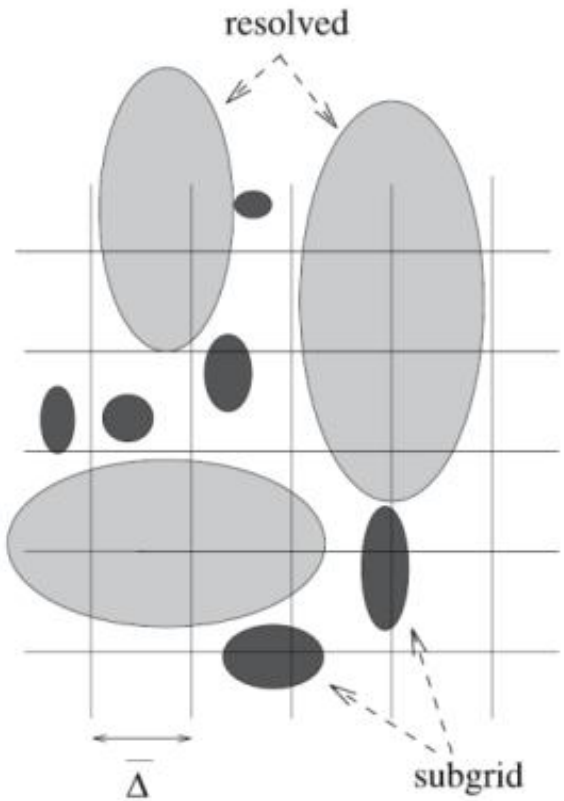
Reynolds Averaged Navier-Stokes (RANS)



Large-Eddy Simulation (LES)



Direct Numerical Simulation (DNS)



Application of a filter to a turbulent flow field in the context of LES

Animation: Laminar Flow

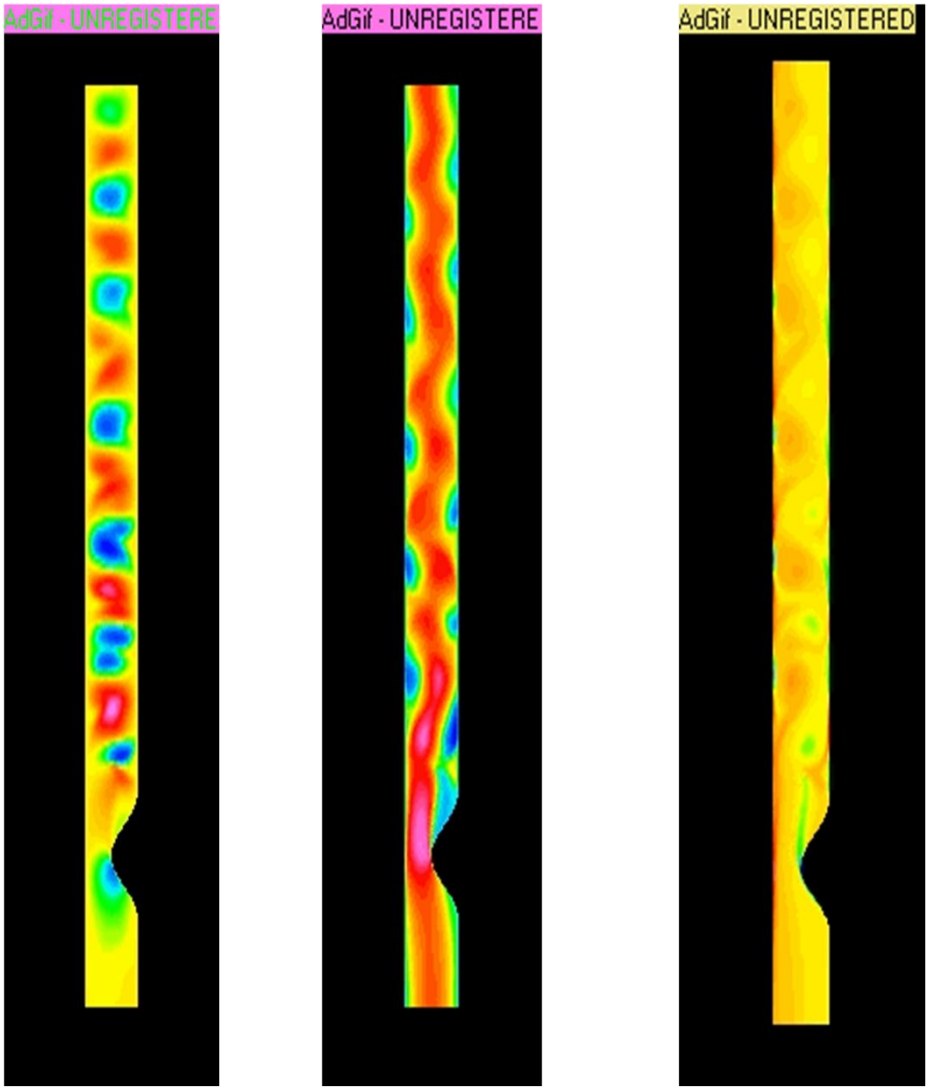


Fig: Sinusoidal pulsatile flow for Re = 500

Animation: Transition to Turbulent Flow

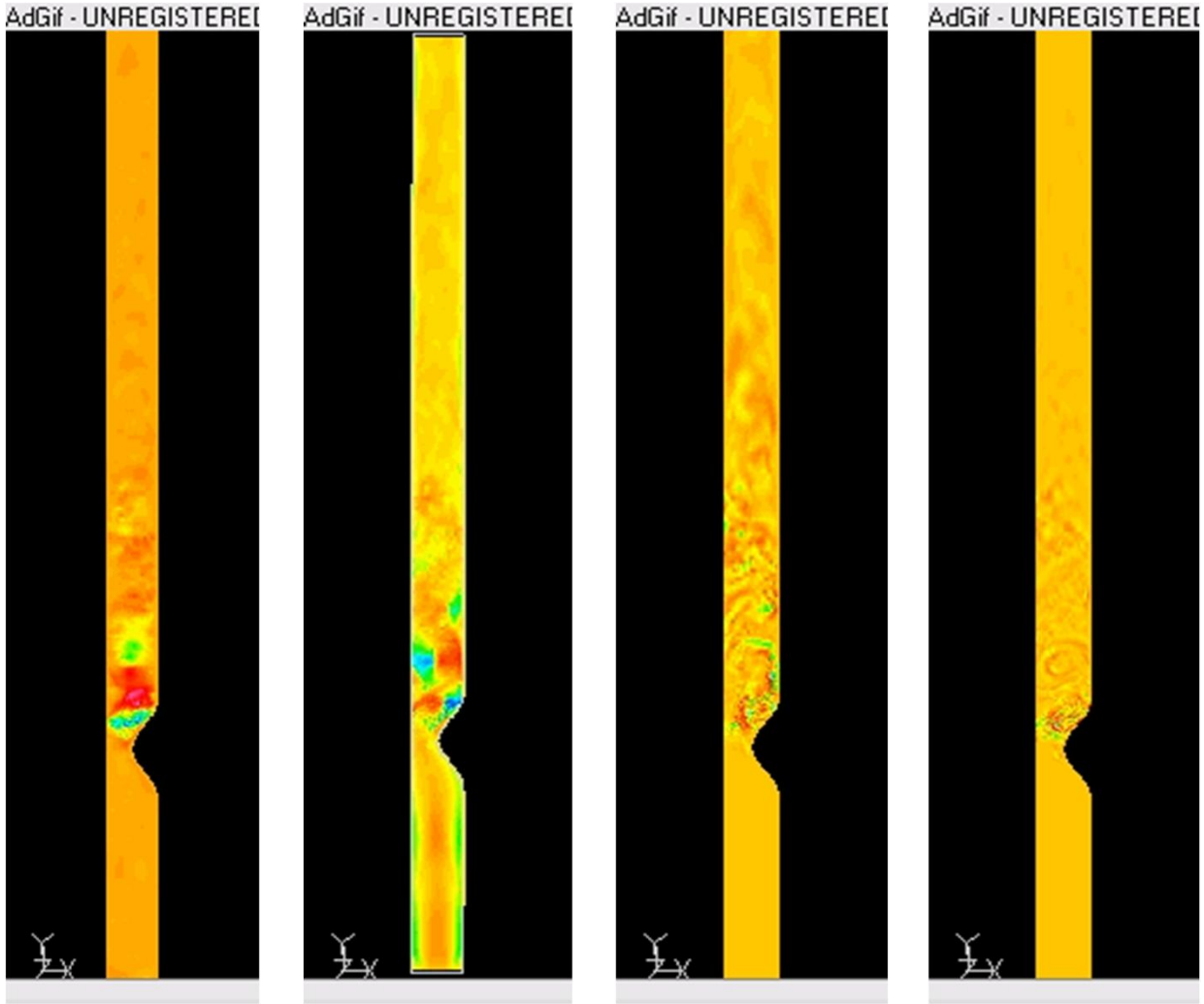
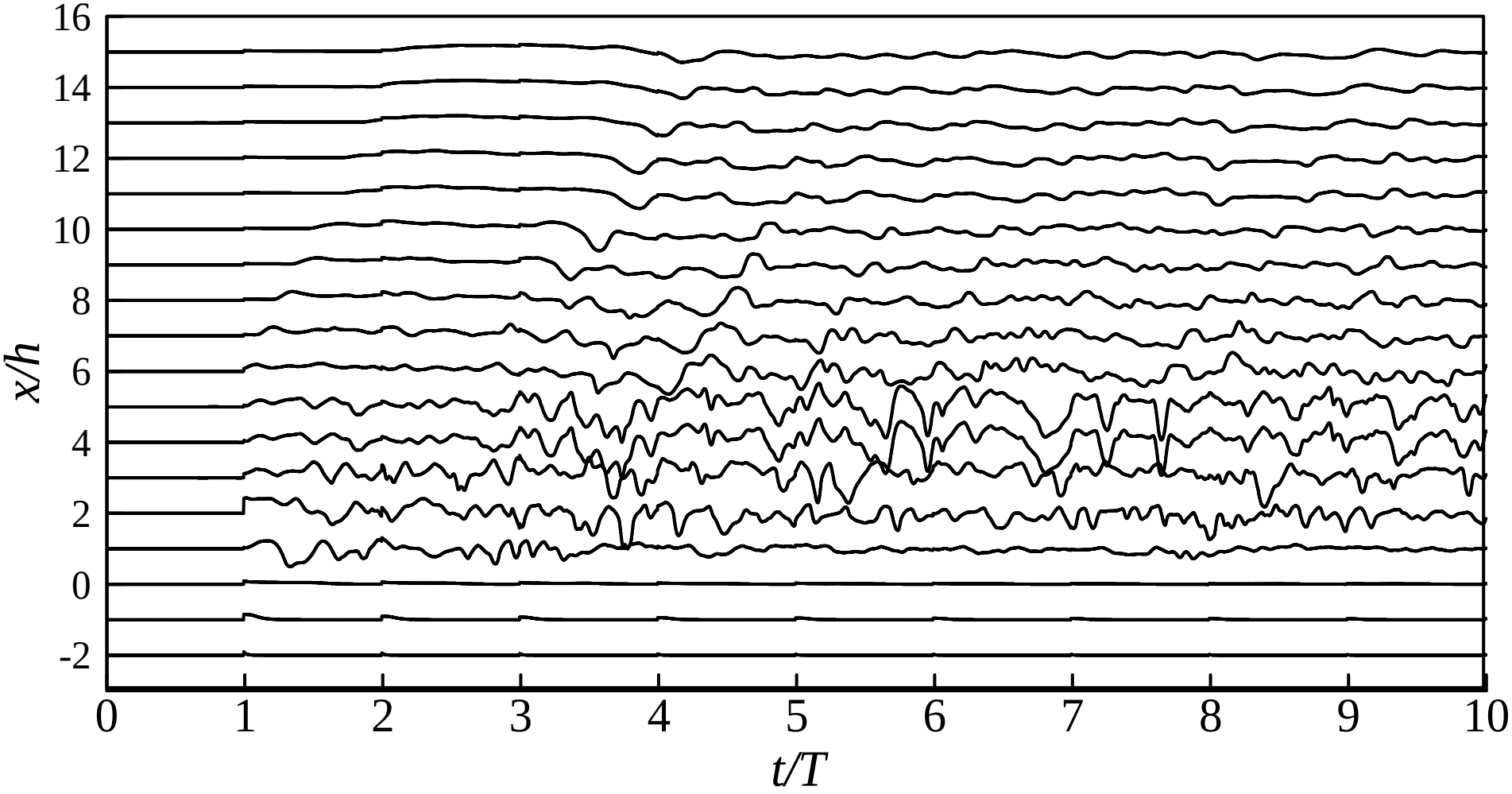


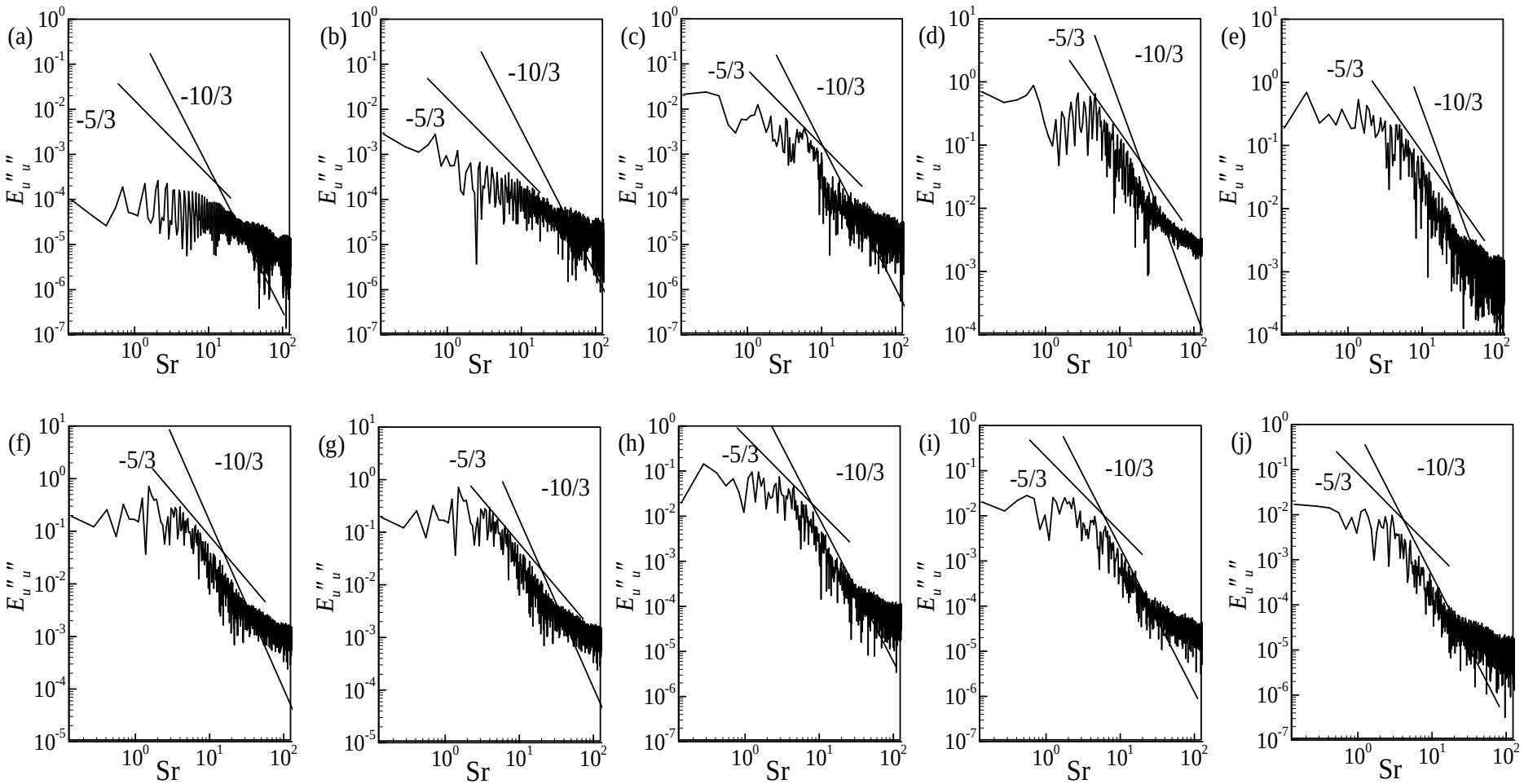
Fig: Physiological Pulsatile flow for Re=2000

Results for Single-Sided Constriction

Time History of the Streamwise Velocity Fluctuations



Energy Spectra for Streamwise Velocity Fluctuations



Energy spectra at (a) x/h = -2 (b) x/h = 0 (c) x/h = 1 (d) x/h = 2 (e) x/h = 3 (f) x/h = 4 (g) x/h = 5 (h) x/h = 6 (i) x/h = 10 and (j) x/h = 14 for *Re* = 1800.